

Multimodal fusion: A study on speech-text emotion recognition with the integration of deep learning

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ABSTRACT

Recognition of various human emotions holds significant value in numerous real-world scenarios. This paper focuses on the multimodal fusion of speech and text for emotion recognition. A 39-dimensional Mel-frequency cepstral coefficient (MFCC) was used as a feature for speech emotion. A 300-dimensional word vector obtained through the Glove algorithm was used as the feature for text emotion. The bidirectional gate recurrent unit (BiGRU) method in deep learning was added for extracting deep features. Subsequently, it was combined with the multi-head self-attention (MHA) mechanism and the improved sparrow search algorithm (ISSA) to obtain the ISSA-BiGRU-MHA method for emotion recognition. It was validated on the IEMOCAP and MELD datasets. It was found that MFCC and Glove word vectors exhibited superior recognition effects as features. Comparisons with the support vector machine and convolutional neural network methods revealed that the ISSA-BiGRU-MHA method demonstrated the highest weighted accuracy and unweighted accuracy. Multimodal fusion achieved weighted accuracies of 76.52 %, 71.84 %, 66.72 %, and 62.12 % on the IEMOCAP, MELD, MOSI, and MOSEI datasets, suggesting better performance than unimodal fusion. These results affirm the reliability of the multimodal fusion recognition method, showing its practical applicability.

1. Introduction

As artificial intelligence rapidly progresses, human-computer interaction technology has witnessed significant progress, and emotion recognition has emerged as a crucial component. Emotions serve as a vital means for humans to convey information. Emotion recognition can help computers perceive human emotions, which has been applied in various fields. For instance, in medical health, emotion recognition facilitates the diagnosis of a patient's condition (Lin et al., 2022). In intelligent driving, driver emotion recognition can be used to provide corresponding driving reminders (Du et al., 2021). In online education, timely recognition of students' emotions helps in understanding the teaching situation (Tanko et al., 2022). As machine learning and deep learning continue to evolve, more and more methods have been applied to emotion recognition (Liu et al., 2021). Zhong et al. (2021) put forward a face emotion recognition approach that combines the histogram of orientation gradient and ensembles with shared representations. It demonstrated high accuracy in experiments on the FER2013 dataset. Polo et al. (2021) analyzed multimodal physiological signals and achieved recognition accuracies of 72 %, 67 %, and 89 % for happiness,

relaxation, and fear, respectively, using a linear discriminant analysis model. Bandela et al. (2021) improved the identification performance of a support vector machine (SVM) through feature selection and a denoising algorithm in a speech emotion recognition study and obtained results surpassing the baseline model. Praveen et al. (2021) proposed an audio-visual (A-V) modality-based emotion recognition method, obtained features through cross-attention fusion, and validated the method through experiments on RECOLA and Fatigue datasets. However, much of the current research predominantly focuses on unimodal features, resulting in low recognition accuracy that falls short of real-life requirements. Multimodal fusion refers to the integration of features from various modalities. It can accomplish recognition tasks based on richer information, and it has been successfully applied in various fields. Wang et al. (2023) proposed a student behavior portrait based on multimodal fusion and used an optimized decision tree-based approach to achieve early warning for abnormal behaviors among college students. Moreover, they verified the accuracy of this approach through experiments. Mou et al. (2023) designed an attention-based multimodal fusion model that integrates features from the eyes, mouth, and optical flow of the head to detect driver fatigue. The experimental results on two datasets

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demonstrated the superiority of this model. Selvi and Vijayakumaran (2023) considered fusing data from three modalities and analyzed their correlations and differences. Experimental findings revealed that this approach achieves higher accuracy in recognizing various emotions. Wang et al. (2021) combined textual and image characteristics extracted from user posts to recommend similar users on social media platforms. The method achieved a maximum average top-k accuracy of 0.491 on the real-world Instagram platform. Current research indicates that multimodal fusion outperforms single modality approaches. Thus, this article proposed a multimodal fusion approach that combines speech and text features. An improved sparrow search algorithm-improved sparrow search algorithm-multi-head self-attention (ISSA-Bi-GRU-MHA) method was designed for emotion recognition. Its reliability was verified. This work presents a novel approach to emotion recognition and offers theoretical references for further research on multimodal fusion.

2. Speech-text emotion features

2.1. Speech emotion features

Speech serves as the primary mode of human communication, including a wealth of emotion-related feature information. In the realm of current research on speech processing, the Mel-frequency cepstrum coefficient (MFCC) stands out as one of the most commonly employed techniques and has demonstrated success in various speech recognition applications (Wibawa & Darmawan, 2021). Therefore, this paper adopts MFCC as the speech emotion feature.

Before extracting the MFCC, the speech signal needs to be pre-processed. First of all, the pre-emphasis can realize the enhancement of the high-frequency part. If the speech signal at the n -th moment is $x(n)$, after pre-emphasis processing, $y(n) = x(n) - \alpha x(n-1)$, $\alpha = 0.98$.

The signal after pre-emphasis is processed by framing and windowing. To obtain a smoother signal, a Hamming window is used. Then, the spectral signal is obtained by fast Fourier transform (FFT): $X(K) = \sum_{n=0}^{N-1} x(n) \cdot e^{-j\frac{2\pi}{N}nk}$, where N is the window length and K is the length of Fourier transform.

The spectral line energy is computed based on the spectrum, and then logarithmic energy $S(i, m)$ is obtained by a set of Mel filters, and then the final MFCC is obtained by discrete cosine transform (DCT):

$$MFCC(i, n) = \sqrt{\frac{2}{M} \sum_{m=0}^{M-1} \log[S(i, m)] \cos\left[\frac{\pi n(2m-1)}{2M}\right]},$$

where M is the number of filters.

After acquiring the MFCC, to increase the practice between the frames, the first- and second-order differences can be added, and the calculation formula can be written as:

$$d_t = \begin{cases} c_{t+1} - c_t, & t < K \\ \frac{\sum_{k=1}^K k(c_{t+k} - c_{t-k})}{\sqrt{2 \sum_{k=1}^K k^2}}, & \text{other} \\ c_t - c_{t-1}, & t \geq Q - K \end{cases}$$

where d_t stands for the t -th first-order difference, c_t is the t -th MFCC, Q stands for the order, and K is the time disparity of the first-order derivative.

The second-order difference parameter can be acquired by substituting the obtained results, and finally, a 39-dimensional MFCC feature is obtained, including:

- (1) 12-dimensional MFCC + one-dimensional energy,
- (2) the first-order difference of 12-dimensional MFCC + the first-order difference of one-dimensional energy,
- (3) the second-order difference of 12-dimensional MFCC + the second-order difference of one-dimensional energy.

2.2. Text emotion features

Indeed, speech inherently carries textual information, which is rich in emotion-related feature information. Textual emotion features primarily involve word vectors. Currently, widely utilized word vector models include Word2Vec (Yu et al., 2023) and Glove (Li et al., 2021). The Glove model, proposed to address shortcomings in the former, is chosen in this paper for obtaining word vector features.

Glove first needs to construct the co-occurrence probability matrix, expressed as:

$$\begin{cases} p_{ij} = p(j|i) = \frac{x_{ij}}{x_i} \\ x_i = \sum_{j=1}^N x_{ij} \end{cases},$$

where p_{ij} denotes the co-occurrence probability of word i and j in the context, x_{ij} is the occurrence probability of word j in the context of the word i , and x_i is the total amount of words appearing in the context of word i .

In a word list with a size of V , given word k that may appear around words i and j , the correlation between i, j , and k is determined. Referring to literature (Pennington et al., 2014) and considering word vectors v_i, v_j, v_k , there is:

$$F(i, j, k) = \frac{p_{i,k}}{p_{j,k}} = g(v_i, v_j, v_k),$$

where $p_{i,k}$ indicates the co-occurrence probability of words i and k in the context and $p_{j,k}$ is the co-occurrence probability of words j and k in the context.

A loss function is constructed:

$$J = \sum_{i,j=1}^V f(x_{i,j}) [v_i^T v_j + b_i + b_j - \log(x_{i,j})]^2,$$

where b_i and b_j are bias terms and $f(x_{i,j})$ is a weighting function.

$$f(x) = \begin{cases} (x/x_{max})^a, & x < x_{max} \\ 1, & \text{otherwise} \end{cases}.$$

The value of x_{max} is set as 100 based on experience, and the value of a is set as 3/4. Each word can eventually be transformed into a 300-dimensional word vector through the Glove model.

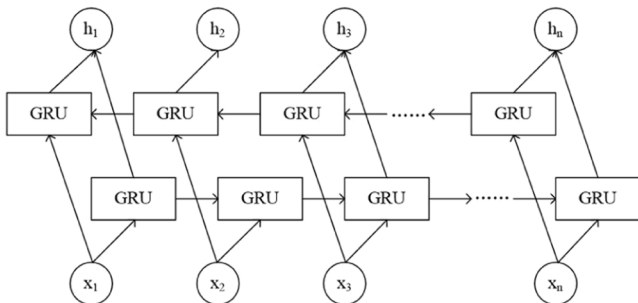


Fig. 1. BiGRU structure.

3. Deep learning-based emotion recognition method

3.1. Bidirectional gated recurrent unit

Speech and text emotion feature data are both sequence data, and the utilization of a recurrent neural network (RNN) (Spencer et al., 2021), a deep learning method, can well capture the temporal dependency in them. Gate recurrent unit (GRU) is an improved method for RNN and long and short-term memory neural network (LSTM), which can not only solve the long-term dependency in RNN, but also simplify the structure of the model to enhance the training speed, and it has been widely used in various recognition fields (Geng et al., 2019). Compared with LSTM, GRU controls the delivery of feature information through only two gated units.

BiGRU is a method designed based on GRU that can consider both forward and backward information at the same time step (Ayasi et al., 2022), to better capture contextual information in emotional features of speech text. Its structure is displayed in Fig. 1.

As shown in Fig. 1, BiGRU consists of a forward GRU and a backward GRU, which process features from front to back and back to front, respectively, and the final output of BiGRU can be obtained by stitching the outputs of each time step.

3.2. Multimodal fusion-based emotion recognition method

In this paper, emotion recognition is considered in terms of both speech and text. In multimodal emotion recognition, let the speech emotion feature be x_A and the text emotion feature be x_T . x_A and x_T are spliced to be the input of BiGRU, and the deep emotion features are obtained after learning. To obtain more significant features, this paper combines the MHA mechanism with BiGRU (Rendón-Segador et al., 2021) to further optimize the features. Moreover, an ISSA is developed to optimize the hyperparameters of BiGRU.

In the multi-head attention, Q , K , and V denote the Query matrix, Key matrix, and Value matrix, respectively, and the attention score is obtained by the Softmax function. The formula can be written as:

$$\text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right),$$

where $\sqrt{d_k}$ is a scaling factor and d_k is a vector dimension.

The above equation is multiplied with the Value matrix to get the final attention matrix:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V.$$

Multi-head attention is formed by the interaction of multiple attentions, which can pay attention to different positional information, and the formula can be written as:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O,$$

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V),$$

where W_i^Q , W_i^K , and W_i^V are weight matrices and W^O is a matrix to be learned.

In the ISSA, the specific improvements for the SSA are as follows.

(1) The cubic chaotic mapping is used to improve the population initialization method in order to enhance the global search capability of the algorithm. The expression of cubic is:

$$x_{n+1} = ax_n(1 - x_n^2),$$

where x_n is the chaotic variable value of the current iteration, 0.2 here, and a is the system parameter, 0.2493 here.

(2) Non-inertial weights are used to optimize the location update

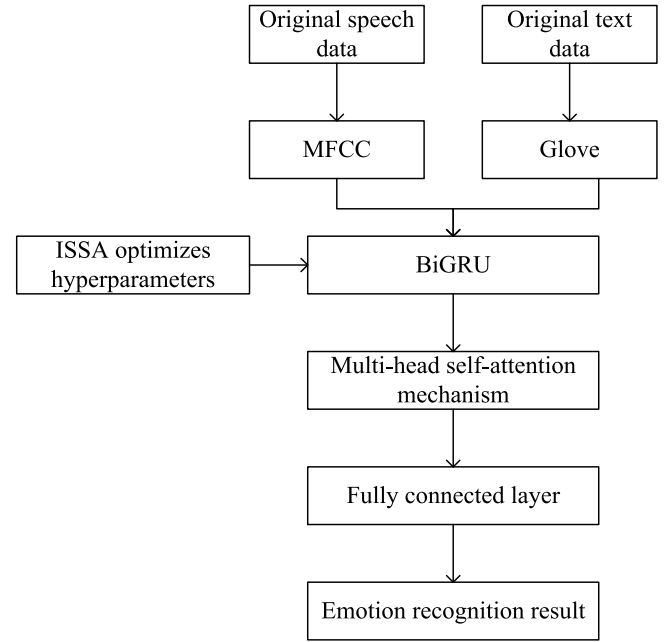


Fig. 2. ISSA-BiGRU-MHA.

formulas for discoverers and sentinels. The new update formula of the discoverer's location is:

$$X_i^{t+1} = \begin{cases} X_i^t \cdot \exp\left(\frac{-i}{\alpha \cdot t_{max}}\right), R_2 < ST \\ w_1 \cdot X_i^t + Q \cdot L, R_2 \geq ST \end{cases}$$

$$w_1 = \sin\left[m \cdot \left(\frac{t}{t_{max}}\right)^r\right],$$

where X_i^t is the location of individual i in the t -th iteration, α is a random number, $\alpha \in (0, 1)$, R_2 is the early warning value, $R_2 \in [0, 1]$, ST is the safety value, $ST \in [0.5, 1]$, Q is a random value obeying normal distribution, L is a $1 \times d$ matrix composing of elements equaling to 1, r and m are adjustment parameters of adaptive inertial weight w_1 , $r = 3$, $m = \pi/2$.

The new update formulas of the sentinel's location are:

$$X_i^{t+1} = \begin{cases} w_2 \cdot X_b^t + \beta \cdot |X_i^t - X_b^t| \cdot f_i > f_g \\ w_1 = \sin\left[mw_1 = \sin\left[m \cdot \left(\frac{t}{t_{max}}\right)^r\right] \left(\frac{t}{t_{max}}\right)^r\right] \end{cases}$$

$$w_2 = 1 - \cos\left[n \cdot \left(\frac{t}{t_{max}}\right)^r\right],$$

where X_w^t is the worst location in the t -th iteration, X_b^t is the best location in the t -th iteration, β is a random number obeying normal distribution with an average value of 0 and a variance of 1, f_i is the current fitness value, f_g and f_w are the optimal and worst fitness values, K is a random number, $K \in [-1, 1]$, ε is a constant that tends to zero infinitely, and n is an adjustment parameter of adaptive inertial weight w_2 , $n = \pi/3$.

The final obtained method combining BiGRU, ISSA, and the multi-head attention mechanism is ISSA-BiGRU-MHA (Fig. 2).

The specific process of the BiGRU-MHA multimodal fusion emotion recognition method is as follows.

- (1) MFCC and Glove word vectors are extracted from the original speech and text data, respectively.
- (2) BiGRU is initialized, and the ISSA is employed to obtain the optimal hyperparameters.

Table 1
Experimental datasets.

	Training	Validation	Testing
IEMOCAP	3641	522	1041
MELD	7919	862	2116
MOSI	1284	299	686
MOSEI	16,326	1871	4659

Table 2
Confusion matrix.

		Identification result	
		Positive	Negative
True label	Positive	TP	FN
	Negative	FP	TN

- (3) Speech and text emotion features are input into the ISSA-BiGRU to learn to get deep features.
- (4) The output features of the ISSA-BiGRU are fed into the MHA mechanism to learn to obtain more salient features.
- (5) After inputting the features output from the previous layer into the fully connected layer, a linear transformation is performed, and then the result is input into the softmax classifier to get the final emotion recognition result.

4. Results and analysis

4.1. Experimental setup

The experiments were implemented in a Windows 10 environment, utilizing the PyTorch framework as the foundation, and implemented in the Python programming language. In the BiGRU-MHA model, the input features were a 39-dimensional MFCC and a 300-dimensional word vector. After optimization by the ISSA, the batch size was 24, the learning rate was set at 0.006, and the dropout rate was 0.5. The Adam optimization algorithm and the cross-entropy loss function were employed for training.

The IEMOCAP (Busso et al., 2008), MELD (Poria et al., 2018), MOSI (Zadeh et al., 2016), and MOSEI datasets (Abdusalomov et al., 2023) were selected to evaluate the proposed method, both of which contain speech and text data and emotion labels. Table 1 shows the data distribution.

The IEMOCAP and MELD datasets each contain four emotions: happiness, appreciation, anger, and neutrality. The MOSI and MOSEI datasets include seven emotions: highly negative, negative, weakly negative, neutral, weakly positive, positive, and highly positive.

The performance of the suggested method was judged by employing the confusion matrix (Table 2).

- (1) Weighted accuracy (WA)

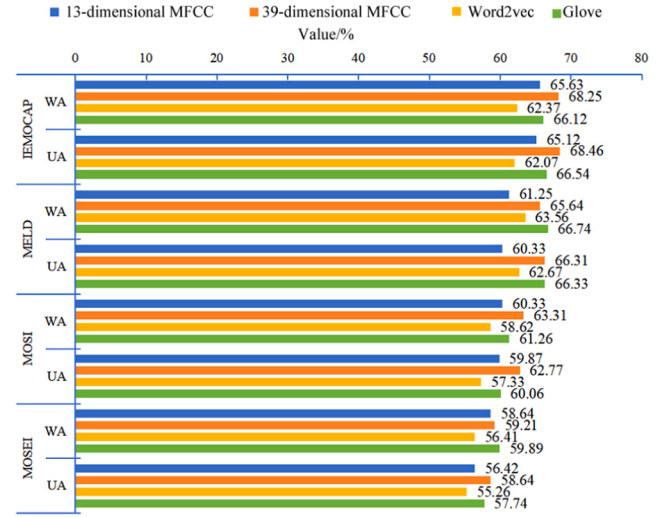
$$WA = \frac{\sum_{i=1}^N TP_i}{\sum_{i=1}^N (TP_i + FP_i)}$$

- (2) Unweighted accuracy (UA)

$$UA = \left[\sum_{i=1}^N \frac{TP_i}{(TP_i + FP_i)} \right] / N$$

4.2. Experimental results

In the extraction of speech emotion features, a comparison was made

**Fig. 3.** Comparison of different feature extraction methods.**Table 3**
Identification results for IEMOCAP (%).

	Speech unimodal		Text unimodal		Speech-text multimodal fusion	
	WA	UA	WA	UA	WA	UA
SVM	32.36	32.27	55.61	55.22	64.25	64.13
XGBoost	52.84	52.21	56.93	55.24	66.77	66.05
DNN	55.33	55.26	57.33	57.02	68.94	68.65
CNN	57.76	57.57	58.64	58.52	70.12	70.07
RNN	59.36	59.08	60.33	60.01	71.36	71.05
LSTM	60.45	60.33	61.27	60.84	72.49	72.13
GRU	62.34	62.16	62.36	62.03	73.55	73.46
BiGRU	64.84	64.25	64.72	64.21	74.21	74.08
CAN	66.77	66.25	64.97	64.56	74.98	75.33
AMFBP	67.24	67.14	65.43	65.21	75.29	76.12
ISSA-BiGRU-MHA	68.25	68.46	66.12	66.54	76.52	77.13

between the recognition effects of the 13-dimensional MFCC and the 39-dimensional MFCC. Additionally, in the extraction of text emotion features, the recognition effects of the Word2Vec model and the Glove model were compared. The results are presented in Fig. 3.

In Fig. 3, it is observed that in speech emotion feature extraction, when utilizing only the 13-dimensional MFCC, the WA of the ISSA-BiGRU-MHA method was 65.63 %, 61.25 %, 60.33 %, and 58.64 %, respectively, for the four datasets. When using the 39-dimensional MFCC, the WA of this method improved to 68.25 %, 65.64 %, 60.33 %, and 58.64 %, marking an enhancement of 2.62 %, 4.39 %, 60.33 %, and 58.64 % compared to the 13-dimensional MFCC, respectively. This outcome suggested that both the first- and second-order differences in MFCC contained valuable information for emotion recognition. The utilization of the 39-dimensional MFCC proved more effective in adequately explaining speech emotion features.

In the context of textual emotion feature extraction, when using the word vector extracted by the Word2Vec model as the feature, the BiGRU-MHA method achieved a WA of 62.37 %, 63.56 %, 58.62 %, and 56.41 % for the four datasets, respectively. However, when using the word vector extracted by the Glove model, the ISSA-BiGRU-MHA method exhibited a WA of 66.12 %, 66.74 %, 61.26 %, and 59.89 %, reflecting an enhancement of 3.75 %, 3.18 %, 2.64 %, and 3.48 %. These results suggested the reliability of the Glove model for extracting text emotion features.

The ISSA-BiGRU-MHA method was compared with conventional machine learning and deep learning methods, including SVM (Kumar et al., 2019), XGBoost (Zhou & Lai, 2023), deep neural network (DNN)

Table 4
Identification results for MELD (%).

	Speech unimodal		Text unimodal		Speech-text multimodal fusion	
	WA	UA	WA	UA	WA	UA
SVM	32.74	32.21	52.62	53.13	64.25	64.31
XGBoost	48.52	49.21	59.03	58.54	65.77	65.56
DNN	49.34	49.02	60.33	60.12	65.92	65.67
CNN	50.33	50.12	61.27	61.05	66.03	66.12
RNN	50.57	50.64	62.35	62.33	66.77	66.27
LSTM	51.21	52.23	63.56	63.35	68.34	68.04
GRU	53.56	54.01	64.01	65.98	68.77	68.35
BiGRU	58.64	58.32	65.12	65.07	69.45	68.59
BiGRU-MHA	65.64	66.31	66.74	66.33	71.84	69.12

(Chen & Yan, 2023), convolutional neural network (CNN) (Kotucha, 2021), RNN, LSTM, GRU, BiGRU, cross attention network (Lee et al., 2020), adaptive and multi-level factorized bilinear pooling (Zhou et al., 2021), and ISSA-BiGRU-MHA.

Table 3 presents the results obtained for the IEMOCAP dataset.

According to Table 4, the performance of traditional machine learning methods such as SVM and XGBoost in emotion recognition was average. The result indicated that traditional machine learning methods generally performed averagely in extracting emotional features, whether it was in single-modal recognition of speech text or multi-modal fusion recognition.

In the comparison of several deep learning methods, it is observed that firstly, the WA and UA of the DNN and CNN methods showed only marginal improvement compared to the SVM and XGBoost methods. However, the recognition effect of the RNN method surpassed that of the DNN and CNN methods, suggesting superior performance in processing sequence data. Moving on to the comparison of various RNNs, the LSTM and GRU approaches are both improvements on the traditional RNN, resulting in higher WA and UA in both unimodal and multimodal recognition. Specifically, the GRU method exhibited a WA of 62.34 % in speech unimodal recognition, 62.36 % in text unimodal recognition, and 73.55 % in speech-text multimodal fusion recognition. These results outperformed the LSTM method, affirming the reliability of GRU in emotion recognition.

In the final comparison of the GRU, BiGRU, CAN, AMFBP, and ISSA-BiGRU-MHA methods, in the context of speech unimodal recognition, for instance, the BiGRU approach exhibited a 2.5 % improvement in WA compared to the GRU method. Furthermore, the ISSA-BiGRU-MHA method demonstrated an additional 3.41 % improvement in WA compared to the BiGRU method and improvements of 1.48 % and 1.01 % compared to the CAN and AMFBP methods. These results demonstrated the effectiveness of utilizing BiGRU for feature extraction, combining it with MHA to further optimize the features, and optimizing hyperparameters using the ISSA.

In the comparison between unimodal and multimodal approaches, employing ISSA-BiGRU-MHA as the recognition method, it is observed that the WA and UA of speech and text unimodal recognition fell below 70 %. This result suggested that unimodal emotion features alone were insufficient to comprehensively explain different emotion types. However, when speech-text multimodal fusion was employed for recognition, the WA reached 76.52 %, and the UA reached 77.13 %, indicating the superiority of multimodal fusion in emotion recognition.

The results obtained for the MELD dataset are displayed in Table 4.

As shown in Table 4, similar to the IEMOCAP dataset, the SVM and XGBoost methods were less effective in emotion recognition. On the other hand, the deep learning methods not only demonstrated the ability to extract deeper features from speech and emotion data but also considered the temporal relationship between sequences comprehensively. Consequently, the deep learning methods achieved better results in emotion recognition.

In the comparison of different deep learning methods, the GRU

Table 5
The identification results obtained for the MOSI and MOSEI datasets (unit: %).

	MOSI		MOSEI	
	WA	UA	WA	UA
SVM	58.91	56.44	54.02	53.17
XGBoost	59.02	57.11	54.68	53.89
DNN	60.21	58.79	55.13	54.26
CNN	61.87	59.33	55.78	54.97
RNN	62.02	60.76	56.71	55.16
LSTM	63.21	61.28	57.33	56.37
GRU	64.33	62.77	58.26	58.27
BiGRU	65.41	63.01	59.84	59.84
CAN	65.97	63.89	60.75	60.88
AMFBP	66.08	64.07	61.87	61.03
ISSA-BiGRU-MHA	66.72	64.33	62.12	61.74

exhibited superior performance compared to the CNN, DNN, and other approaches. For instance, in the case of speech unimodal recognition, the WA of the GRU was 53.56 %, and its UA was 54.01 %, surpassing the performance of the CNN, DNN, and similar methods. Notably, the BiGRU method showed further improvement over the GRU method, with a 5.08 % increase in WA and a 4.31 % increase in UA. The ISSA-BiGRU-MHA method outperformed the BiGRU method with a 7 % improvement in WA and a 7.99 % improvement in UA. Moreover, it also showed improvements in WA and UA compared to the CAN and AMFBP methods. These results once again verified the advantage of the ISSA-BiGRU-MHA method in emotion recognition.

For the MELD dataset, speech-text multimodal fusion recognition demonstrated superior performance compared to unimodal recognition. The ISSA-BiGRU-MHA method achieved a WA of 71.84 % in multimodal fusion recognition, showing a 6.2 % improvement compared to speech unimodal and a 5.1 % improvement compared to text unimodal. The results further verified the reliability of multimodal fusion in emotion recognition.

Table 5 presents the WA and UA obtained for the MOSI and MOSEI datasets using different methods under multimodal fusion.

It can be seen that, similar to the IEMOCAP and MELD datasets, machine learning methods exhibited poor emotion recognition performance on the MOSI and MOSEI datasets. Even with multimodal fusion, both WA and UA were below 60 %. However, there were improvements in the performance of deep learning methods. Comparatively, the proposed ISSA-BiGRU-MHA model also demonstrated the highest WA and UA on the MOSI and MOSEI datasets, further confirming its advantage in emotion recognition.

5. Conclusion

This paper introduces deep learning into speech-text multimodal fusion for emotion recognition and proposes the ISSA-BiGRU-MHA method. Experimental results on the IEMOCAP, MELD, MOSI, and MOSEI datasets revealed that the MFCC speech emotion features and Glove text emotion features utilized in this study yielded superior recognition effects. Furthermore, the recognition performance of multimodal fusion surpassed that of unimodal recognition significantly. The proposed method exhibits distinct superiority in emotion recognition compared to methods such as SVM and CNN, suggesting its potential for practical applications and widespread adoption.

Declaration of competing interest

The authors declare they have no conflicts of interests.

Data availability

Data will be made available on request.

Credit Author Statement

I confirm that the paper entitled Multimodal fusion: a study on speech-text emotion recognition with the integration of deep learning has never been submitted or published on any other journal before.

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